Transfer Learning and Featurization for Classification of Eye Diseases on Retinal Images

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Background

Diabetic retinopathy is present in over 18.2 million people in America

In 2020, there were around 237.1 million people with moderate or severe visual impairments caused by cataracts.

Glaucoma is expected to rise from 76 million in 2020 to 118.1 million in 2040

These diseases have no cure but can be prevented from further progression if diagnosed early.

Unfortunately, many do not have access to sufficient eye care due to a lack of efficient and accurate diagnosis in their area. If not diagnosed and managed in early stages, **glaucoma, diabetic retinopathy, and cataracts** can lead to severe visual impairment or blindness, lowering patients' quality of lives.

Introduction

Conventionally, these diseases are **detected manually** by ophthalmologists. However, human ophthalmologists are prone to misdiagnosis, are more expensive, and take more time for diagnosis.

Patient goes for regular eye screening/has noticed visual degradation. → Physician potentially misdiagnoses patient. → Patient is unable to manage disease in time, resulting in its further progression.

ARTIFICIAL INTELLIGENCE

Automation through artificial intelligence (AI) techniques has been extensively explored in recent years. AI tools have the potential to perform **complex tasks** extremely **efficiently**, helping **diagnosis** become more **accessible** and **inexpensive**.

Deep learning and **machine learning** have been used for a variety of image classification tasks including disease diagnosis. In this study, we explore two AI techniques, **transfer learning** and **featurization**, for the diagnosis and classification of eye diseases in retinal images.

Goals

RATIONALE: A lack of access to efficient and inexpensive eye diseases diagnosis has resulted in the progression of such illnesses, leading to severe visual impairment and blindness among patients. AI can solve this by providing a low-cost, rapid, accurate detection tool that analyzes retinal scans for certain diseases.

RESEARCH QUESTION: Can transfer learning and featurization create AI models that will accurately diagnose eye diseases?

HYPOTHESIS: Transfer learning and featurization will be able to create a model with high accuracy value that can rapidly diagnose whether one has cataracts, glaucoma, diabetic retinopathy, or normal eyes from their retinal image.

ENGINEERING GOAL: To create an AI model that utilizes transfer learning and/or featurization for the diagnosis of glaucoma, diabetic retinopathy, and cataracts by analyzing retinal images.

MATERIALS:

- Eye Disease Classification Dataset
- Google Colab
- MobileNetV2, ResNet50, VGG16 pre-trained models
- KNN, Random Forest, MLP Alghorithms
- Google Drive

Methodology and Experimental Details

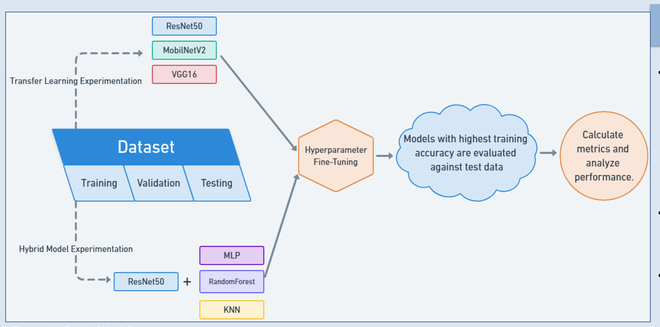

Figure 1. Entire Workflow

Dataset

- retinal images collected from several sources such as Eyepacs, Indian Diabetic Retinopathy Dataset, Digital Retinal Images for Vessel Extraction, Ocular Recognition, and High-Resolution Fundus Image Database.
- contains a total of 4217 retinal images of healthy eyes and diseased eyes with GC, DR, and cataracts
- Dataset split into train, validation, and test datasets in the proportions of 80%, 10%, and 10%.

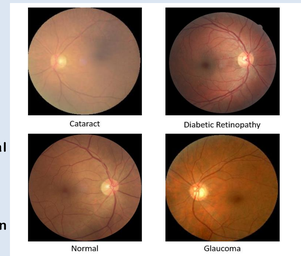

Figure 2. Retinal Images from the dataset

Transfer Learning

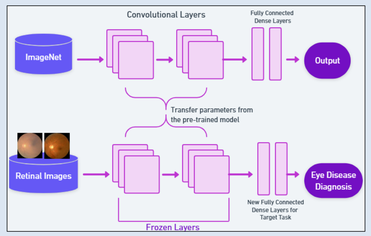

Figure 3. Transfer Learning Schematic

- To each pre-trained model, 2 Dense layers were added (1st: 100 neurons, 2nd: 4 neurons)
- Pre-trained models: MobileNetV2, VGG16, ResNet50
- Different hyperparameters were tested for each model.
- Models with the highest training accuracy were evaluated against test data.

Featurization

- Since ResNet50 performed the best out of the 3 transfer learning models, it was used for featurization
- 2048 features were extracted
- Features were used by KNN, Random Forest, and Multi-Layer Perceptron (MLP) classifiers

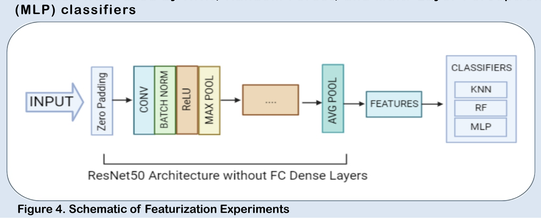

Figure 4. Schematic of Featurization Experiments

Statistical Analysis

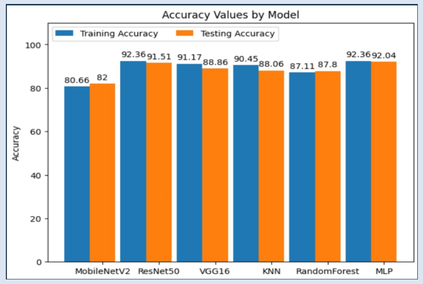

Figure 12. All accuracy values of models

- **ResNet50** (50 epochs; 0.0001 learning rate): highest accuracy out of all transfer learning experiments
  - Training accuracy: 92.4%
  - Testing accuracy: 91.5%
- **ResNet50 + MLP Classifier** (20 epochs; 0.001 learning rate): highest accuracy out of all experiments
  - Training accuracy: 92.4%
  - Testing accuracy: 92.04%

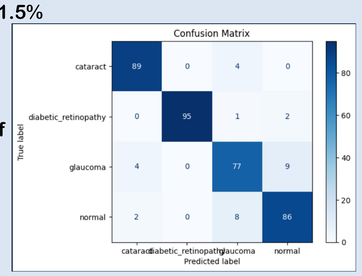

Figure 13. Confusion Matrix for MLP

Key Experimental Results

Transfer Learning

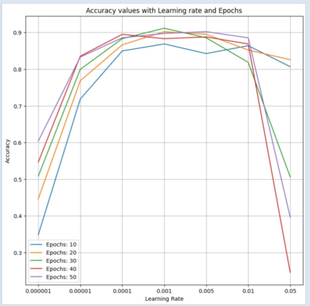

Figure 5. Accuracy Graph for VGG16

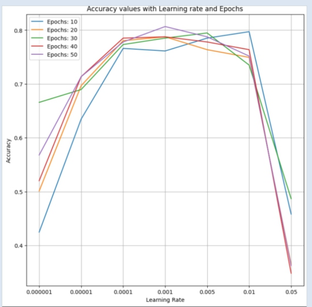

Figure 6. Accuracy Graph for MobileNetV2

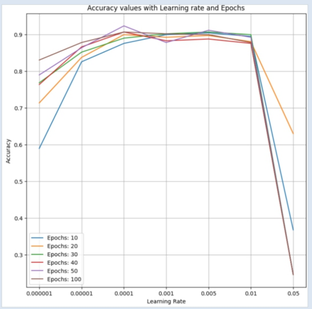

Figure 7. Accuracy Graph for ResNet50

Featurization

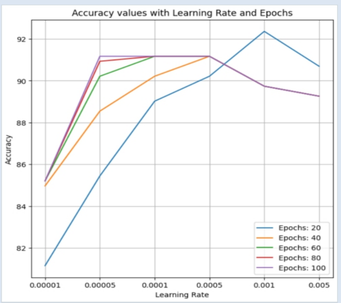

Figure 8. Accuracy Graph for MLP

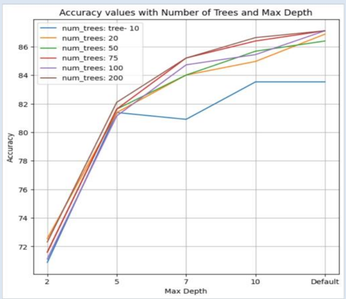

Figure 9. Accuracy Graph for Random Forest

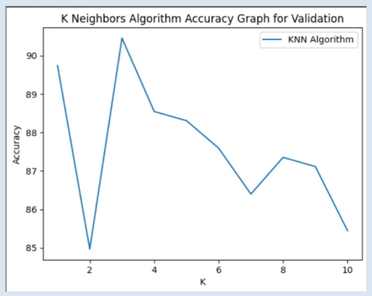

Figure 10. Accuracy Graph for KNN

Table 1. Classification Report for MLP Classifier

Label	Precision	Recall	F1-Score
Cataract	93.2%	93.2%	93.2%
Diabetic Retinopathy	97.3%	99.1%	98.2%
Glaucoma	95.3%	81%	87.6%
Normal	85%	95.3%	89.9%

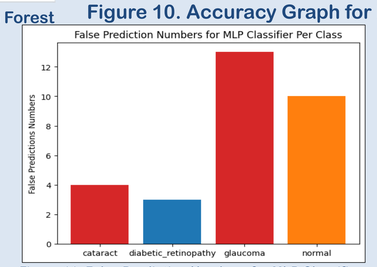

Figure 11. False Prediction Numbers for MLP Classifier

Conclusion

Out of all techniques tested, the model that implemented featurization with ResNet50 and combined it with MLP Classifier diagnosed eye images with the highest accuracy.

Advantages of this approach:

- High Accuracy
- Time Efficient
- No Associated Costs
- Increased Accessibility

Final decisions regarding treatment will always be made by an experienced ophthalmologist, however, AI diagnostic tools make this process more efficient and inexpensive, ultimately increasing its accessibility and helping patients manage eye diseases in a timely manner.

Key References

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